

3. A retailer of satellite dishes would like to know the impact of advertising on her sales. For 6 months she records the number of ads run in the newspaper and the number of sales:

Worksheet: Adsales.mtw

	Mar	Apr	May	Jun	Jly	Aug
Number of ads	0	3	5	8	10	9
Number of sales	5	7	12	15	17	15

Regression Analysis

The regression equation is
 $\text{sales} = 4.67 + 1.23 \text{ ads}$

Predictor	Coef	SE Coef	T	p
Constant	4.6748	0.8018	5.83	0.004
ads	1.2272	0.1176	10.44	0.000

$s = 1.017$

$R\text{-sq} = 96.5\%$

$R\text{-sq}(\text{adj}) = 95.6\%$

- Draw the scatterplot of the data. Does there appear to be a linear trend in the data?
- Test the hypothesis that $\beta_1 = 0$. Does the result of this test indicate that the linear model is a reasonable model? What is the value of the coefficient of determination?
- Predict the number of sales if the retailer runs six ads in June.
- With the additional information, $SS_x = 74.83$ and $\bar{x} = 5.83$, construct a 98% confidence interval for the expected sales when six ads are placed.
- Obtain the deletion residual corresponding to Aug observation.